

CHASSERAL, Switzerland

WMO: 066050

Lat: 47.1317N

Lon: 7.0544E

Elev: 5351

StdP: 12.07

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.4	11.1	-16.3	2.8	30.7	-8.8	4.2	28.7	87.1	29.5	77.0	30.3	20.8	60	0.985

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	70.6	58.5	67.8	57.8	65.2	56.5	60.7	67.4	59.0	65.4	57.5	63.5	11.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.0	87.8	64.2	56.4	82.8	62.4	54.9	78.3	60.7	29.6	67.7	28.3	65.6	27.2	63.7	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	48.8	41.2	DB	6.7	74.5	3.6	3.0	4.1	76.6	2.0	78.3	0.0	80.0	-2.6	82.1
			WB	5.2	64.4	3.5	1.6	2.7	65.5	0.6	66.5	-1.3	67.4	-3.8	68.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	27.1	27.0	31.1	38.0	43.4	51.4	54.5	54.0	47.7	43.2	34.3	29.9	(6)
	DBStd	12.59	7.75	8.20	8.18	8.05	7.76	7.72	7.02	7.29	7.06	7.92	8.48	8.24	(7)
	HDD50	4143	711	643	586	366	232	77	35	41	126	230	472	624	(8)
	HDD65	9064	1176	1063	1051	810	668	412	331	349	519	675	921	1089	(9)
	CDD50	570	0	0	0	6	29	118	174	165	57	20	1	0	(10)
	CDD65	15	0	0	0	0	0	2	5	8	0	0	0	0	(11)
	CDH74	11	0	0	0	0	0	0	7	4	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
	WSAvg	19.4	23.7	23.3	20.9	17.5	17.6	16.3	15.8	15.6	16.3	19.6	21.9	24.3	(14)
	PrecAvg	47.7	3.5	3.4	3.1	3.6	4.4	4.3	4.5	4.6	3.9	4.1	4.1	4.3	(15)
(16) Precipitation	PrecMax	61.8	8.9	8.1	10.9	7.0	8.1	8.7	9.5	10.8	7.2	7.2	9.9	9.3	(16)
	PrecMin	34.3	0.8	1.0	0.9	0.8	2.0	2.3	2.0	0.9	1.5	0.7	0.4	0.1	(17)
	PrecStd	7.0	2.1	1.8	2.3	1.9	1.8	1.7	1.7	2.0	1.7	1.7	2.3	2.2	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	49.3	50.8	59.5	65.4	72.2	74.6	74.0	64.9	61.3	55.2	48.6	(19)
		MCWB	33.5	34.4	39.4	48.3	55.0	60.2	59.9	58.1	55.7	49.6	41.8	34.6	(20)
	2%	DB	43.0	44.3	47.3	54.9	61.5	68.6	70.1	70.6	62.1	58.0	51.0	45.8	(21)
		MCWB	33.0	33.2	38.1	45.0	52.0	58.4	58.5	58.1	54.3	47.6	40.3	33.5	(22)
	5%	DB	40.1	40.8	44.4	52.1	58.0	65.7	67.2	67.7	59.7	55.5	48.2	43.3	(23)
		MCWB	33.1	31.9	36.3	43.5	49.8	56.9	57.5	57.5	52.9	46.5	39.8	33.2	(24)
	10%	DB	37.4	37.5	42.0	49.2	54.8	62.9	64.5	64.5	57.5	53.1	45.6	40.4	(25)
		MCWB	32.4	31.4	35.3	41.8	47.8	55.1	56.2	56.6	51.7	45.5	39.0	33.3	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	38.1	41.9	49.0	56.5	62.8	62.5	62.6	58.7	52.5	46.4	40.4	(27)
		MCDB	41.7	41.1	46.3	56.7	63.9	69.6	70.2	69.5	62.4	56.4	50.6	42.2	(28)
	2%	WB	37.5	35.8	40.2	46.4	53.3	60.1	60.2	60.3	56.0	50.6	43.6	38.3	(29)
		MCDB	39.2	38.9	44.5	53.0	59.2	66.4	67.3	67.1	59.8	54.2	46.9	40.3	(30)
	5%	WB	35.6	34.2	38.5	44.4	50.9	58.0	58.6	58.6	54.4	49.1	42.0	36.8	(31)
		MCDB	37.1	37.9	42.5	50.5	56.4	64.1	65.2	64.9	58.3	52.8	45.3	39.5	(32)
	10%	WB	33.4	32.6	36.6	42.6	48.6	55.9	57.1	57.0	52.5	47.5	40.4	35.2	(33)
		MCDB	36.1	36.6	40.5	48.4	53.8	61.4	63.2	63.1	56.5	51.1	43.9	38.2	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	8.0	7.6	8.2	9.6	10.2	10.4	10.7	9.7	8.8	8.6	7.9	8.1	(35)
		MCDBR	10.2	10.1	10.3	12.4	13.4	12.4	12.9	11.4	10.7	10.2	10.1	10.3	(36)
		MCWBR	6.8	6.6	7.2	8.8	9.6	10.1	9.4	9.3	9.5	8.1	6.9	6.2	(37)
	5% WB	MCDBR	8.9	9.2	9.3	11.9	12.6	12.1	12.0	11.2	10.0	8.9	8.8	8.5	(38)
		MCWBR	8.3	7.6	7.2	8.8	9.4	10.3	9.2	9.4	9.4	7.9	7.6	7.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.249	0.268	0.294	0.321	0.329	0.332	0.331	0.326	0.310	0.294	0.269	0.247	(40)
	taud		2.519	2.487	2.477	2.421	2.430	2.446	2.434	2.463	2.498	2.540	2.559	2.522	(41)
	Ebn at Noon		277	292	298	297	297	295	294	292	288	277	266	264	(42)
	Edh at Noon		19	24	29	34	35	35	35	32	28	23	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg		376	641	1020	1381	1594	1887	1822	1584	1216	771	418	305	(44)
	RadStd		47	99	154	203	151	175	161	147	113	75	53	55	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (108 neighbors)	N/A	N/A	N/A	+1.31	+0.72	+0.54	N/A	N/A	+95	+29

Nomenclature: See separate page